

Dualformer Controllable Fast And Slow Thinking By Learning With Randomized Reasoning Traces

1. One-Line Summary

Dualformer trains a single Transformer on multiple compressed versions of reasoning traces, enabling it to perform both fast answer generation and slow explicit reasoning while often outperforming Searchformer.

2. Problem

Existing approaches face a tradeoff:

- Fast Mode: direct answer generation, efficient but less accurate.
 - Slow Mode: explicit reasoning traces, more accurate but computationally expensive. Prior work such as Searchformer and Beyond A* Search requires long traces and iterative bootstrapping to obtain shorter reasoning processes.
- The key question is:
Can a single Transformer learn both fast and slow reasoning without additional controllers, multiple models, or iterative retraining?

3. Core Idea

Instead of always training on complete reasoning traces, randomly remove structured portions of the trace during training.

The same problem is therefore observed at multiple reasoning granularities:

- Full reasoning
- Partially compressed reasoning
- Highly compressed reasoning
- No reasoning

The model learns full reasoning depths rather than a single fixed reasoning style

4. Key Concepts

Fast Mode

Problem → Answer. No explicit reasoning generated.

Slow Mode

Problem → Reasoning Trace → Answer

Explicit search or reasoning process generated.

Auto Mode

The model chooses reasoning depth automatically.

Structured Trace Dropping

Three dropping operations:

- D1: Remove close clauses.
- D2: Remove cost tokens.
- D3: Remove some create clauses.

Level Hierarchy

Level 1:

- D1

Level 2:

- D1 + D2

Level 3:

- D1 + D2 + D3

Level 4:

- Remove entire trace.

Randomized Training

For every training instance, a dropping level is sampled randomly.

The same task appears with different reasoning depths throughout training (as we repeat this for 1000 times).

5. Important Equations

No major theoretical derivations are introduced. Purely a new training algorithm.

Problem + Trace + Solution $\rightarrow$ Problem + Compressed Trace + Solution

where compression level is sampled during training.

The mathematics section later uses reasoning-step dropping probability p :

Reasoning Step $\rightarrow$ dropped with probability p for synthetic math reasoning traces.

6. Method Pipeline

Training

1. Generate Searchformer-style traces.
2. Apply one of the dropping levels.
3. Train using standard autoregressive next-token prediction.
4. Repeat with randomized dropping.

Inference

Fast Mode:

concatenate 'plan' to the prompt

Generates answer directly.

Slow Mode:

concatenate 'create' to the prompt

Generates reasoning trace and answer.

Auto Mode:

No control token.

The model decides how much reasoning to perform.

7. Results

- In fast mode Dualformer significantly outperforms Solution-Only models. Eg: (30×30 Maze): Dualformer ~80% and Solution-Only ~30%.
- Slow Mode outperforms Searchformer while generating much shorter reasoning traces. 1538 vs 854 tokens.
- Auto Mode achieves near Slow-Mode accuracy with substantially reduced computation. More reasoning used on harder mazes.
- Diversity of generated solutions increases compared to Searchformer.
- Math reasoning experiments show small but consistent improvements, suggesting the idea transfers beyond planning.

8. Weaknesses

- Generalization to unseen maze distributions is limited.(0% in 0.1 ,0 .2).
- Improvements on math reasoning tasks are almost nil.
- The paper demonstrates that trace dropping works but does not fully explain why.
- System 1/System 2 is mainly a motivating analogy rather than a proven mechanism.

9. My Thoughts

- Why does dropping reasoning information improve performance?
- Even this training is done over 1000 iterations then how are they claiming searchformer is more expensive.
- The main contribution is not the specific dropping operations, but exposing the model to multiple levels of reasoning granularity.

10. Connections

- **Searchformer:** Extends Searchformer by training on compressed reasoning traces instead of only full traces.
- **Beyond A*:** Learns compressed reasoning directly rather than through iterative bootstrapping.
- **Chain-of-Thought Distillation:** Uses reasoning traces as supervision to improve direct answering.
- **Dropout:** Similar intuition of removing information during training to improve robustness and generalization.